

Project Title

Metro Ticket Generating System in ServiceNow

Name : Lodagala Maneesha

College : Aditya Institute of Technology and Management

Email : 23a51a0536@adityatekkali.edu.in

Project Report

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Project Overview

The **Metro Ticket Generating System in ServiceNow** is designed to simplify and digitize the metro ticket booking process. The system enables passengers to book metro tickets through a self-service portal by selecting the source station, destination station, ticket type, and number of passengers.

Once the booking is submitted, ServiceNow processes the request automatically, calculates the fare, creates a booking record, and generates a unique QR code. The generated QR code acts as a digital metro ticket, allowing passengers to travel without purchasing physical tickets at metro stations.

The solution uses various ServiceNow components such as the **Service Portal, Service Catalog, Widgets, Client Scripts, Server Scripts, Script Includes, Business Rules, and Flow Designer** to automate the entire booking lifecycle. The system improves user convenience by providing instant ticket generation, faster processing, secure digital tickets, and reduced manual effort.

This project demonstrates how ServiceNow can be used beyond IT Service Management by automating real-world transportation services. It enhances customer experience, reduces waiting time at ticket counters, minimizes paper usage, and provides a modern, efficient, and user-friendly metro ticket booking system.

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1. Introduction :

In today's fast-paced urban environment, commuters expect a quick, convenient, and reliable method for purchasing metro tickets. Traditional ticketing systems often require passengers to wait in long queues at ticket counters or vending machines, resulting in delays, increased manual effort, and reduced customer satisfaction. During peak hours, these challenges become even more significant, affecting the overall travel experience.

As the number of metro passengers continues to grow, manual ticketing methods become less efficient. Paper-based tickets can be lost or damaged, while manual verification increases operational workload and slows passenger movement at entry gates. These issues highlight the need for a modern, digital, and automated ticketing solution.

The **Metro Ticket Generation System** developed using the **ServiceNow** platform provides a centralized and automated solution for booking metro tickets online. Passengers can easily book tickets through the **ServiceNow Service Portal** by selecting the source station, destination station, journey type, and number of passengers. Once the booking is completed, the system automatically generates a **digital ticket with a unique QR code**, which can be scanned at metro stations for quick and secure entry.

The proposed system offers the following benefits:

- Faster and paperless metro ticket booking
- Digital ticket generation with unique QR codes
- Reduced waiting time at ticket counters
- Secure and accurate ticket validation
- Improved passenger convenience and travel experience
- Automated booking and notification process
- Reduced manual effort and operational costs
- Easy access to booking history through the Service Portal
- Environment-friendly solution by reducing paper usage

By utilizing the powerful capabilities of **ServiceNow**, including the **Service Catalog, Service Portal, Flow Designer, Business Rules, UI Pages, and Widgets**, this project demonstrates how workflow automation can be applied to public transportation services. The Metro Ticket Generation System provides an efficient, secure, and user-friendly digital ticketing solution while supporting future

enhancements such as online payment integration, ticket cancellation, and real-time metro schedule updates.

2. Project Objectives :

The main objective of this project is to develop an automated **Metro Ticket Generation System** using the **ServiceNow** platform that simplifies the metro ticket booking process and provides passengers with a fast, secure, and paperless ticketing experience. The system is designed to offer users a centralized platform for booking metro tickets while automating ticket generation, QR code creation, notifications, and booking management.

The project also focuses on developing a standardized workflow that improves passenger convenience, reduces manual effort, minimizes waiting time at ticket counters, and ensures secure digital ticket validation with minimal human intervention.

The specific objectives of this project are:

- Develop a centralized **Service Catalog** for booking metro tickets.
- Design interactive ticket booking forms using **Catalog Variables** to capture journey details such as source station, destination station, ticket type, and number of passengers.
- Implement **UI Policies** to dynamically display or hide fields based on user selections.
- Create **Custom Tables** to securely store and manage metro ticket booking records.
- Configure **Flow Designer** to automate the complete ticket booking and generation process.
- Automatically generate a unique **QR Code** for every successfully booked metro ticket.
- Send automated **email notifications** containing ticket details and QR codes after successful booking.
- Maintain detailed booking records for monitoring, tracking, and future reference.
- Reduce ticket booking time while eliminating the need for paper tickets and long waiting queues.
- Build a scalable and flexible solution that can be extended to support features such as online payment integration, ticket cancellation, fare calculation, return journey booking, and real-time metro schedule updates in the future.

3. Key Features

- User-friendly **Service Portal** for metro ticket booking.
- Dynamic ticket booking forms with intelligent field visibility.
- **Catalog Variables** for capturing passenger and journey details.
- **Variable Sets** for collecting reusable passenger information.
- Automated ticket booking workflow using **Flow Designer**.
- Custom **Metro Ticket** database table for storing booking records.
- Automatic fare calculation based on source and destination stations.
- Automatic generation of a unique **QR Code** for every booked ticket.
- Digital metro ticket generation with booking details.
- Email notifications containing ticket confirmation and QR code.
- Real-time booking status tracking.
- Secure storage of booking history and passenger records.

- Paperless ticket validation through QR code scanning.
- Scalable workflow architecture for future enhancements such as online payment, ticket cancellation, and return journey booking.
- Reduced ticket booking time and improved passenger convenience.

4. Project Milestones

The project implementation follows five major milestones.

Milestone 1: Custom Table Creation

Objective

Create a custom table to store metro ticket booking details.

Activities:

- Analyze the data required for metro ticket booking.
 - Create a custom **Metro Ticket** table.
 - Create fields for passenger and journey details.
 - Configure field types and properties.
 - Save and verify the custom table.
 - Ensure the table is ready to store booking records.
-

Milestone 2: Service Catalog Creation

Objective

Develop a Service Catalog item for metro ticket booking.

Activities:

- Create a **Service Catalog** item named **Book a Metro Ticket**.
 - Create Catalog Variables for passenger and journey details.
 - Configure variable types and mandatory fields.
 - Arrange variables in a user-friendly format.
 - Apply Catalog Client Scripts and UI Policies (if required).
 - Save and test the catalog item.
-

Milestone 3: Workflow Automation using Flow Designer

Objective

Automate the metro ticket generation process.

Activities:

- Create a new Flow in **Flow Designer**.
 - Configure the **Catalog Item Requested** trigger.
 - Add the **Get Catalog Variables** action.
 - Add the **Create Record** action.
 - Map catalog variables to the Metro Ticket table.
 - Activate the Flow.
 - Verify automatic record creation after ticket booking.
-

Milestone 4: QR Code Integration and Testing

Objective

Generate QR codes and validate the complete ticket booking process.

Activities:

- Create a **UI Page** for displaying the QR code.
 - Integrate the QR Code generation API.
 - Configure the QR popup on the request page.
 - Test QR code generation.
 - Perform unit testing.
 - Perform catalog item testing.
 - Perform flow testing.
 - Verify record creation in the custom table.
 - Conduct end-to-end booking testing.
 - Perform User Acceptance Testing (UAT).
-

Milestone 5: Deployment and Validation

Objective

Deploy the Metro Ticket Generation System and validate its functionality.

Activities:

- Validate all configurations.

- Publish the Service Catalog item.
- Activate the Flow Designer workflow.
- Verify metro ticket generation.
- Verify QR code display and scanning.
- Monitor request processing.
- Train end users.
- Collect feedback and monitor system performance.
- Document the project for future enhancements.

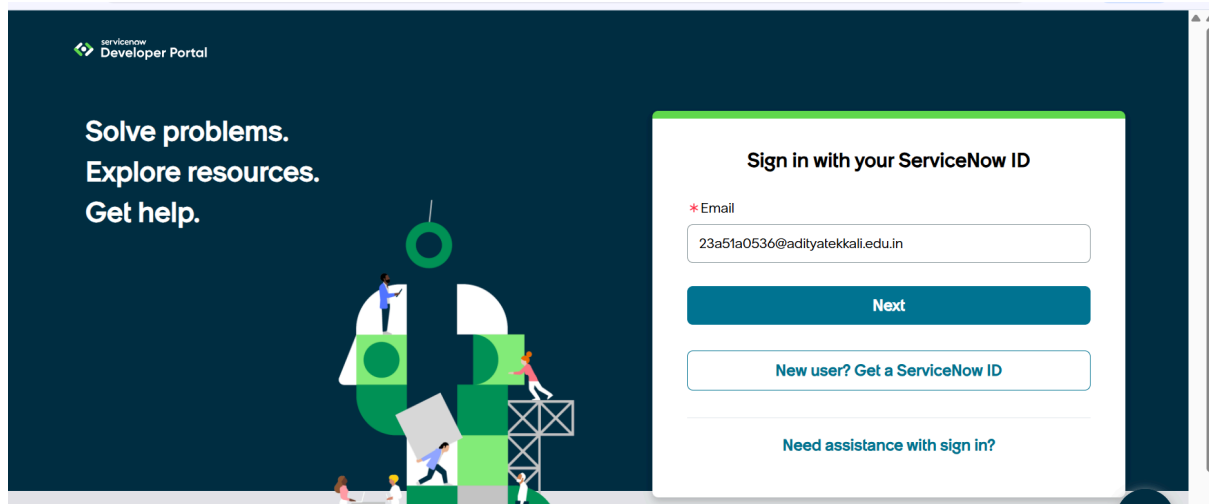
5. ServiceNow Developer Setup

Before developing the application, a ServiceNow Personal Developer Instance (PDI) must be created. The PDI provides a fully functional ServiceNow environment where applications can be developed, tested, and deployed without affecting production systems.

5.1 Creating a ServiceNow Developer Account :

To begin development, perform the following steps:

1. Visit the ServiceNow Developer Portal.
2. Click **Sign Up** to create a free developer account.
3. Enter the required information:
 - First Name
 - Last Name
 - Email Address
 - Password
4. Verify the email address using the activation link sent to your registered email.
5. Log in to the ServiceNow Developer Portal.
6. Click **Start Building**.
7. Request a **Personal Developer Instance (PDI)**.
8. Wait for the instance to be provisioned.
9. Launch the instance and log in using the generated credentials.



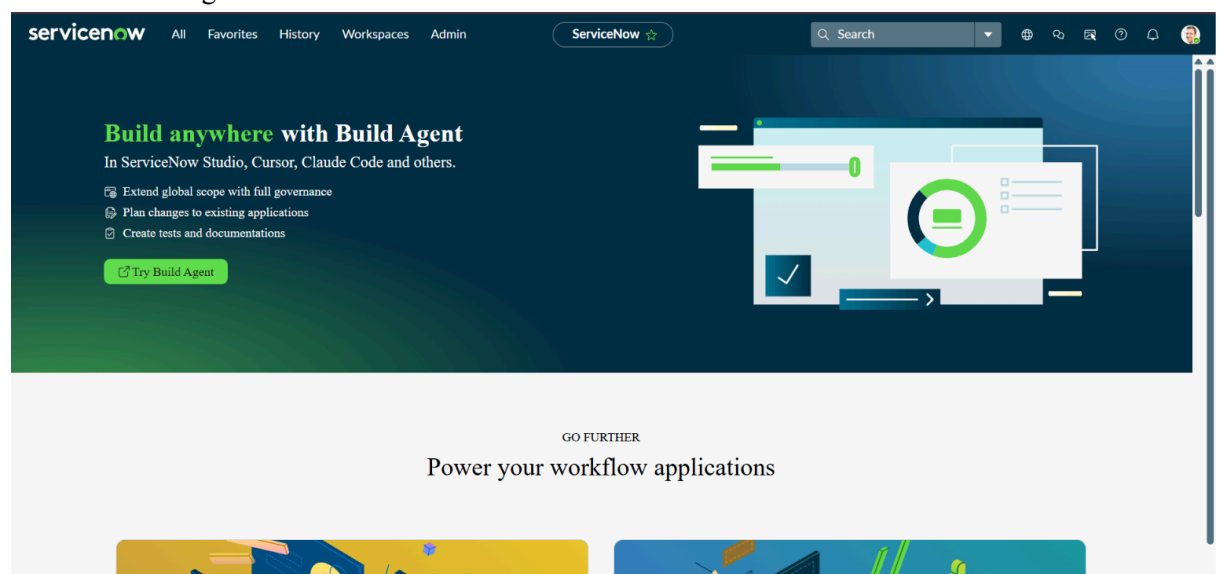
5.2 Accessing Creator Studio

Once the Personal Developer Instance is ready, ServiceNow redirects the developer to Creator Studio.

Creator Studio provides a no-code and low-code environment for building applications. It simplifies application development by allowing developers to create forms, workflows, tables, automation, and integrations through an intuitive graphical interface.

Creator Studio offers features such as:

- Application Creation
- Data Modeling
- Flow Automation
- Form Builder
- Catalog Item Development
- Workflow Configuration



6.1 Custom Table Creation

The **Custom Table** is created to store all metro ticket booking information in a centralized database. It acts as the backend storage for passenger details, journey information, fare details, and ticket records generated through the Service Catalog. Every time a user books a metro ticket, the booking details are automatically stored in this custom table using **Flow Designer**.

Steps to Create the Custom Table

1. Open the **Application Navigator** in the ServiceNow platform.
2. Search for **Tables**.
3. Navigate to the following path:

System Definition → Tables

4. Click the **New** button to create a new table.
5. Enter the required details:
 - **Label:** Metro Ticket
 - **Name:** *(Auto-generated by ServiceNow)*
 - **Application:** Metro Ticket Generation System
6. Click **Submit** to create the custom table.
7. Open the newly created table and add the required fields.

The screenshot displays the ServiceNow interface for creating a custom table. The main form includes fields for 'Label' (Metro Station's Details), 'Name' (u_metro_station_s_details), and 'Extends table'. On the right, there are checkboxes for 'Create module', 'Create mobile module', 'Remote Table', and 'Is Data Interface'. Below the configuration fields, there are tabs for 'Columns', 'Controls', and 'Application Access'. The 'Columns' tab is active, showing a 'Table Columns' section with a search bar and a 'Dictionary Entries' table.

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)		32	false

Purpose of the Custom Table

The purpose of creating the custom table is to:

- Store metro ticket booking records securely.

- Maintain passenger and journey details.
- Keep track of ticket bookings for future reference.
- Support automated ticket generation and QR code creation.
- Enable easy retrieval, reporting, and management of booking records.

Fields Created in the Custom Table

Field Name	Field Type	Purpose
Passenger Name	String	Stores the name of the passenger.
Starting From	Reference/Choice	Stores the source metro station.
Going To	Reference/Choice	Stores the destination metro station.
Journey Type	Choice	Indicates whether the ticket is Single Journey or Return Journey.
Number of Passengers	Integer	Stores the total number of passengers.
Travel Date	Date	Stores the journey date.
Fare Amount	Currency/Decimal	Stores the calculated ticket fare.
QR Code	String	Stores the generated QR code or QR data.
Booking Status	Choice	Indicates the booking status (Booked, Cancelled, Completed).

Table Metro Ticket						
Dictionary Entries						
	Column label	Type	Reference	Max length	Default value	Display
×	Starting From	Reference	Metro Station's Details	32		false
×	Ticket Number	String	(empty)	40		false
	Created by	String	(empty)	40		false
×	Going To	Reference	Metro Station's Details	32		false
×	QR Code	String	(empty)	160		false
×	no.of passangers	Integer	(empty)	40		false
×	Amount	Currency	(empty)	20		false
①	Sys ID	Sys ID (GUID)	(empty)	32		false
	Created	Date/Time	(empty)	40		false
×	Payment Mode	Choice	(empty)	40		false
×	Journey Type	Choice	(empty)	40		false
	Updated by	String	(empty)	40		false
	Updates	Integer	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
+	Insert a new row...					

6.2 Service Catalog Creation

The **Service Catalog** acts as a centralized self-service portal that enables users to submit network service requests easily. In this project, a catalog item named **Network Request** was created to provide users with a single interface for requesting different network-related services such as VPN access, IP allocation, and device configuration.

Steps to Create the Service Catalog Item

1. Open the **Application Navigator** in the ServiceNow platform.
2. Search for **Service Catalog**.
3. Navigate to the following path:
Service Catalog → Maintain Items
4. Click the **New** button to create a new catalog item.
5. Enter the required details:
 - **Catalog Item Name:** Book a metro ticket
 - **Catalog:** Service Catalog
 - **Category:** Services
 - **Short Description:** a metro e-ticketing system allow passengers to book tickets digitally.
 - Click **Save** to create and store the catalog item successfully.

Catalog Item
Book A Metro Ticket

Build and modify items faster with the improved [Catalog Builder](#).

Catalog Items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name:

Application:

Catalogs:

Category:

State:

Checked out:

Owner:

Secondary owners:

Active: ☒

Fulfillment automation level:

Purpose of the Catalog

The **Service Catalog** serves as the primary interface through which passengers can book metro tickets conveniently. It provides a centralized and user-friendly platform that standardizes the metro ticket booking process, enabling users to submit booking requests quickly through the **Service Portal**.

Using the **Book a Metro Ticket** catalog item, users can:

- Select the **Starting Station**
- Select the **Destination Station**
- Choose the **Journey Type** (Single Journey or Return Journey)
- Enter the **Number of Passengers**
- Select the **Travel Date**
- View the **Calculated Fare**
- Submit the booking request
- Receive a **digital metro ticket with a QR code**

By providing a single point of access for metro ticket booking, the Service Catalog eliminates the need for manual ticket counters, reduces waiting time, improves booking efficiency, and ensures secure, paperless ticket management. The system also enables automatic fare calculation, ticket generation, QR code creation, and easy tracking of booking requests through the ServiceNow platform.

6.3 Catalog Variables Configuration

Catalog Variables are used to collect the necessary information from passengers when they book a metro ticket. These variables appear as input fields on the **Metro Ticket Booking** catalog item and capture the journey details required for ticket generation. The information entered through these

variables is later used by **Flow Designer** to create a booking record in the custom Metro Ticket table and automate the ticket generation process.

Steps to Create Catalog Variables

Follow the steps below to configure catalog variables:

1. Open the **Book a Metro Ticket** catalog item.
2. Navigate to the **Variables** related list.
3. Click the **New** button to add a new variable.
4. Configure the required properties, including:
 - **Variable Type**
 - **Question**
 - **Name**
 - **Order**
 - **Tooltip**
 - **Help Text**
 - **Mandatory Status**
5. Click **Save** to create the variable.
6. Repeat the above procedure until all required catalog variables have been added to the **Book a Metro Ticket** catalog item.

Repeat the above procedure until all required catalog variables have been added to the Network Request catalog item.

Variable Configuration

Variable Name	Variable Type	Purpose
Passenger Name	Single Line Text	Captures the passenger's name.
Starting From	Reference/Select Box	Selects the starting metro station.
Going To	Reference/Select Box	Selects the destination metro station.
Journey Type	Choice	Allows the user to choose Single Journey or Return Journey.
Number of Passengers	Integer	Captures the number of passengers.

Travel Date	Date	Specifies the date of travel.
Fare Amount	Currency	Displays the calculated ticket fare.

Variables (9)	Variable Sets (1)	Catalog UI Policies (1)	Catalog Client Scripts	Available For	Not Available For	Categories (1)	Catalogs (1)	Catalog Data Lookup Definitions	Related Articles	Related Catalog Items
Assigned Topics										
≡ ▼ Order Search 🔗 📄 📄 — Actions on selected rows... New										
Catalog Item = Network Request										
☐	🔍	Type	Question						Order ▲	
		Container Start	Service Details						200	
		Multiple Choice	Is this a new Network or Relocation?						300	
		Single Line Text	If this is a reloaction, please provide ...						310	
		Container Start	Location & Device Types						400	
		Single Line Text	Please provide address here						410	
		Select Box	Types of Devices						420	
		Single Line Text	Provide device details						430	
		Container Start	Additional Information						500	
		Single Line Text	If any, please write here						510	

Types of Variables Used

1. Reference Variable

A **Reference Variable** is used to retrieve data from another ServiceNow table by referencing an existing record.

Example:

Passenger Name → User Table

This enables the system to automatically access user-related information, reducing manual data entry.

2. Choice Variable

A **Choice Variable** allows users to select one option from a predefined list of values.

Example:

Mode of payment

Available options include:

- UPI

- Card
- Cash

3. Single Line Text

This variable type is used to capture short text values entered by the passenger.

Examples:

- Passenger Name
- Mobile Number
- Email Address

Purpose:

Allows passengers to enter basic personal information required for metro ticket booking and communication.

4. Multi-Line Text

A **Multi-Line Text** variable is used whenever passengers need to provide additional information related to their journey.

Example:

- Additional Remarks
- Special Travel Instructions

Purpose:

Allows passengers to provide extra details or special requests, such as assistance requirements or additional information related to the journey. This information helps administrators process the booking more effectively.

Advantages of Catalog Variables

The use of Catalog Variables provides several benefits, including:

- Standardized collection of user information
- Improved accuracy of submitted data
- Simplified workflow automation
- Better reporting and record management
- Reduced manual verification and validation

6.4 Catalog Client Script Creation

The **Catalog Client Script** is used to perform client-side operations on the **Book a Metro Ticket** catalog item. It provides a dynamic and interactive user experience by automatically calculating the

ticket fare based on the selected source station, destination station, journey type, and number of passengers. The script executes instantly without requiring a page refresh.

Purpose of the Catalog Client Script

The Catalog Client Script is used to:

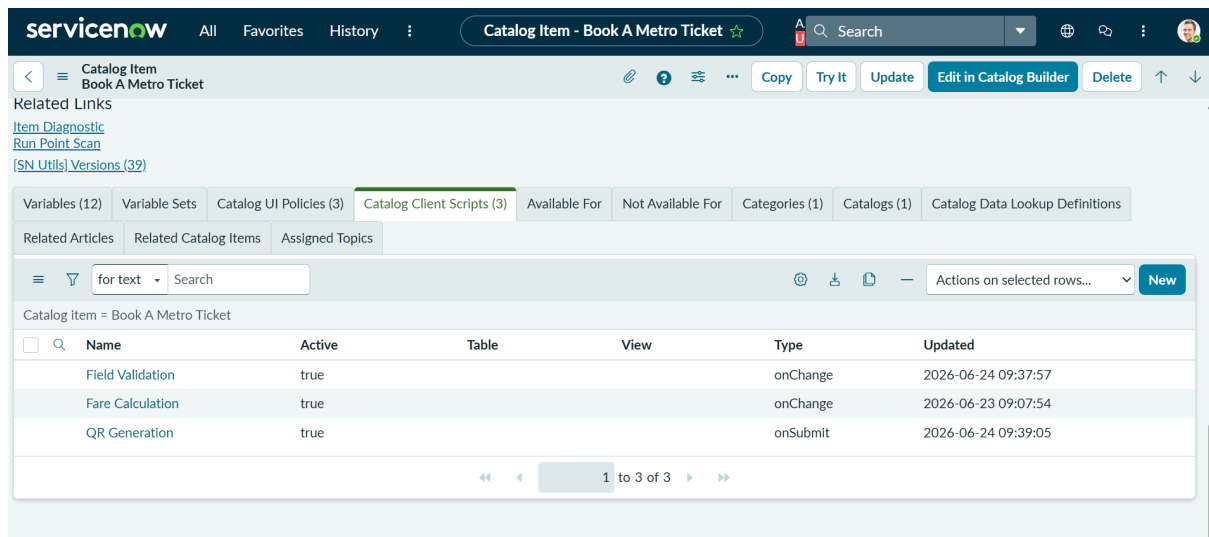
- Dynamically calculate the metro ticket fare.
 - Display the fare automatically based on user selections.
 - Handle both **Single Journey** and **Return Journey** bookings.
 - Improve user experience by providing instant fare calculation.
 - Eliminate manual fare calculation errors.
 - Validate user input before submitting the booking request.
-

Steps to Create the Catalog Client Script

1. Open the **Application Navigator** in the ServiceNow platform.
2. Search for **Catalog Client Scripts**.
3. Navigate to:

Service Catalog → Catalog Definitions → Catalog Client Scripts

4. Click the **New** button.
5. Configure the following details:
 - **Name:** Metro Ticket Fare Calculation
 - **Catalog Item:** Book a Metro Ticket
 - **UI Type:** All
 - **Type:** onChange
 - **Variable Name:** Journey Type (or the variable used to trigger fare calculation)
6. Write the JavaScript code to:
 - Retrieve the selected **Starting Station**.
 - Retrieve the selected **Destination Station**.
 - Retrieve the **Journey Type**.
 - Retrieve the **Number of Passengers**.
 - Calculate the ticket fare.
 - Display the calculated fare in the **Fare Amount** field.
7. Click **Submit** to save the Catalog Client Script.
8. Test the catalog item by selecting different stations and journey types to verify that the fare is calculated correctly.



6.5 Flow Designer Configuration

The **Flow Designer** is used to automate the metro ticket booking process. When a passenger submits the **Book a Metro Ticket** catalog item, the flow is triggered automatically. It retrieves the catalog variables, creates a new record in the **Metro Ticket** custom table, and stores all the booking details. This automation eliminates manual data entry, ensures accurate record creation, and improves the overall ticket booking process.

Purpose of Flow Designer

The Flow Designer is used to:

- Automate the metro ticket booking process.
- Retrieve passenger and journey details from the catalog item.
- Create a booking record in the custom **Metro Ticket** table.
- Eliminate manual record creation.
- Ensure accurate storage of ticket booking information.
- Improve efficiency and reduce processing time.

Steps to Create the Flow

1. Open the **Application Navigator** in the ServiceNow platform.
2. Search for **Flow Designer**.
3. Navigate to:

Process Automation → Flow Designer

4. Click **New** and select **Flow**.
5. Enter the required details:
 - **Flow Name:** Metro Ticket Generation Flow

- **Application:** Metro Ticket Generation System
- 6. Click **Submit**.
- 7. Add the **Trigger**:
 - Select **Service Catalog**.
 - Choose **Catalog Item Requested**.
 - Select the **Book a Metro Ticket** catalog item.
- 8. Add the **Get Catalog Variables** action to retrieve all passenger and journey details.
- 9. Add the **Create Record** action.
- 10. Select the **Metro Ticket** custom table.
- 11. Map the catalog variables to the corresponding fields in the Metro Ticket table.
- 12. Save the flow.
- 13. Click **Activate** to enable the flow.

The top screenshot shows the 'New Flow' setup screen in Workflow Studio. The flow name is 'Metro Project', the application is 'Global', and the description is 'Describe your flow'. The 'Build flow' button is visible at the bottom right.

The bottom screenshot shows the 'Metro Project' flow configuration page. The 'TRIGGER' dropdown menu is open, showing options like 'Record', 'Scheduled', and 'Application'. The 'Application' option is selected, and a sub-menu shows 'Service Catalog' as the chosen item. The 'ACTIONS' section is also visible with 'Add an Action' button.

Flow Actions

- **Trigger:** Catalog Item Requested
- **Action 1:** Get Catalog Variables

- **Action 2:** Create Record
- **Result:** Metro ticket booking details are automatically stored in the custom Metro Ticket table.

After activating the flow, every time a passenger submits the **Book a Metro Ticket** catalog item, the system automatically retrieves the entered details and creates a new booking record in the **Metro Ticket** table, enabling seamless and efficient metro ticket management.

6.5.1 Get Catalog Variables

After the flow is triggered, the next step is to retrieve the information entered by the passenger. This is accomplished using the **Get Catalog Variables** action, which collects all the required input values from the submitted **Book a Metro Ticket** catalog item.

The retrieved data is then used throughout the workflow to create a booking record in the **Metro Ticket** custom table and automate the ticket generation process.

Purpose of Get Catalog Variables

The **Get Catalog Variables** action is used to:

- Retrieve passenger details entered in the catalog item.
- Collect journey information for ticket generation.
- Pass the retrieved values to the **Create Record** action.
- Eliminate manual data entry.
- Ensure accurate processing of metro ticket booking requests.

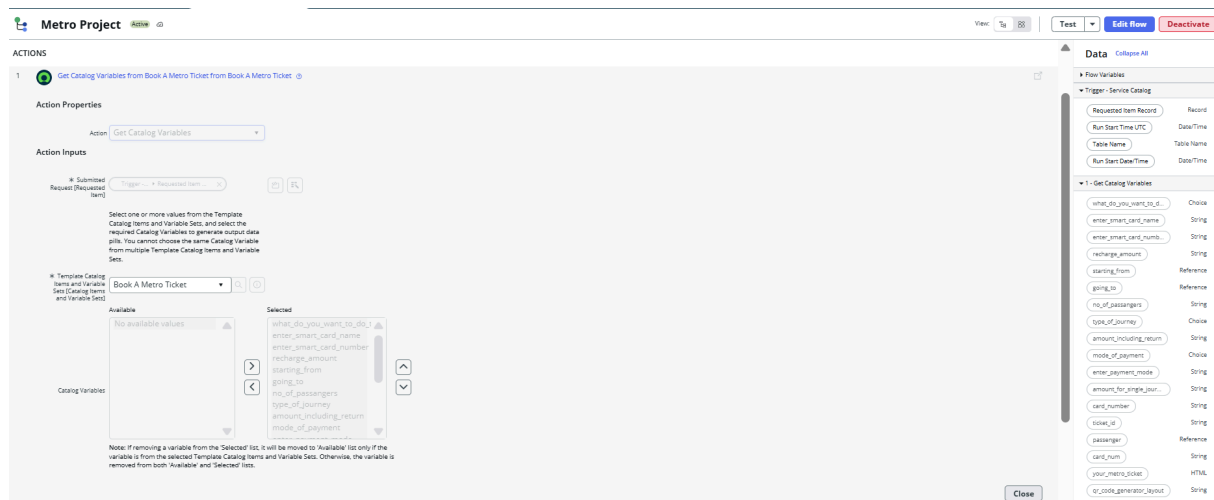
Steps to Configure Get Catalog Variables

1. Open the **Metro Ticket Generation Flow** in **Flow Designer**.
2. Click **Add Action** within the flow.
3. Search for **Get Catalog Variables**.
4. Select the **Get Catalog Variables** action.
5. Configure the **Requested Item** as the input parameter.
6. Select all the required catalog variables.
7. Click **Save** to save the action configuration.

The action retrieves the following booking details:

- Passenger Name
- Starting From
- Going To
- Journey Type
- Number of Passengers
- Travel Date
- Fare Amount

These retrieved values are then passed to the **Create Record** action, which automatically stores the booking details in the **Metro Ticket** custom table. This automation ensures that every metro ticket booking request is processed accurately and efficiently without manual intervention.



6.5.2 Create Record

After retrieving the catalog variables, the next step in the flow is to create a new record in the **Metro Ticket** custom table. This is accomplished using the **Create Record** action in **Flow Designer**. The action maps the retrieved catalog variables to the corresponding fields in the custom table and automatically stores the metro ticket booking information.

Purpose of Create Record

The **Create Record** action is used to:

- Automatically create a new metro ticket booking record.
- Store passenger and journey details in the **Metro Ticket** custom table.
- Eliminate manual data entry.
- Ensure accurate and secure storage of booking information.
- Maintain booking history for future reference and reporting.

Steps to Configure Create Record

1. Open the **Metro Ticket Generation Flow** in **Flow Designer**.
2. Click **Add Action**.
3. Search for **Create Record**.
4. Select the **Create Record** action.
5. Choose the **Metro Ticket** custom table.
6. Map the catalog variables to the corresponding table fields:
 - Passenger Name → Passenger Name
 - Starting From → Starting From
 - Going To → Going To
 - Journey Type → Journey Type

- Number of Passengers → Number of Passengers
 - Travel Date → Travel Date
 - Fare Amount → Fare Amount
7. Click **Save** to save the action configuration.
 8. Click **Activate** to enable the flow.

6.6 Metro Ticket Booking through Service Portal

The **Service Portal** provides a user-friendly interface that allows passengers to book metro tickets online. Users can access the **Book a Metro Ticket** catalog item, enter their journey details, and submit the booking request. After submission, the request is processed automatically through **Flow Designer**, and the booking details are stored in the custom **Metro Ticket** table.

Purpose of the Service Portal

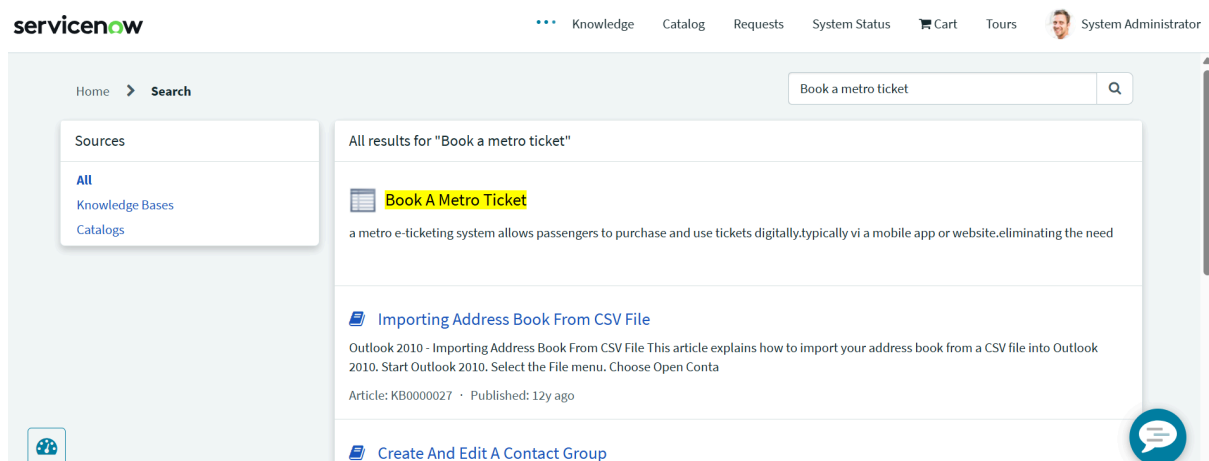
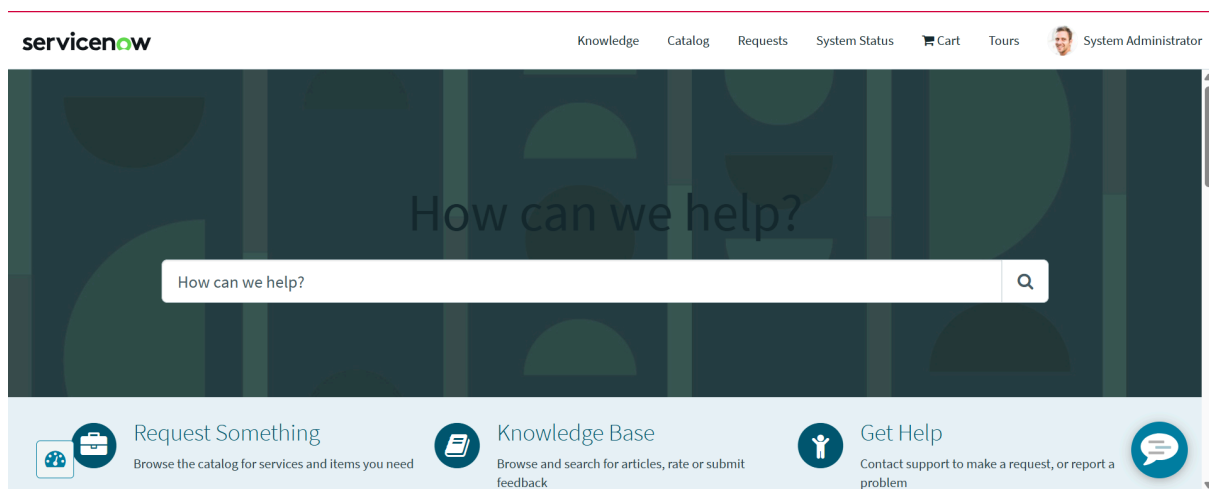
The Service Portal is used to:

- Provide an easy-to-use interface for metro ticket booking.
- Collect passenger and journey details.
- Submit ticket booking requests electronically.
- Trigger the automated workflow for ticket generation.
- Reduce manual effort and improve user experience.

Steps to Book a Metro Ticket

1. Open the **Service Portal** in the ServiceNow platform.
2. Search for **Book a Metro Ticket** in the search bar.
3. Open the **Book a Metro Ticket** catalog item.

4. Fill in the required details:
 - Passenger Name
 - Starting From
 - Going To
 - Journey Type (Single Journey/Return Journey)
 - Number of Passengers
 - Travel Date
5. Verify the automatically calculated fare amount.
6. Click the **Order Now** button to submit the booking request.
7. The system triggers the **Flow Designer** automatically.
8. A new booking record is created in the **Metro Ticket** custom table.
9. The request is successfully submitted, and the passenger can view the generated metro ticket and QR code on the request page.



Home > Service Catalog > Office > Services > **Book A Metro Ticket**

Search Catalog

Single Journey

Amount for single journey?

55

Amount Including Return

Auto calculated

* Mode Of Payment

☐ Net Banking

☐ Others

☒ UPI

Quantity: 1

Delivery Time: 2 Days

Add to Cart

Save as Draft

Order Now

Add attachments

6.7 QR Code Generation

The **QR Code Generation** feature provides passengers with a digital metro ticket after selecting **Book a QR Ticket** and successfully submitting the booking request. The generated QR code contains the unique ticket information, which can be scanned for ticket verification at metro stations. This feature enables a secure, paperless, and contactless ticketing experience.

Purpose

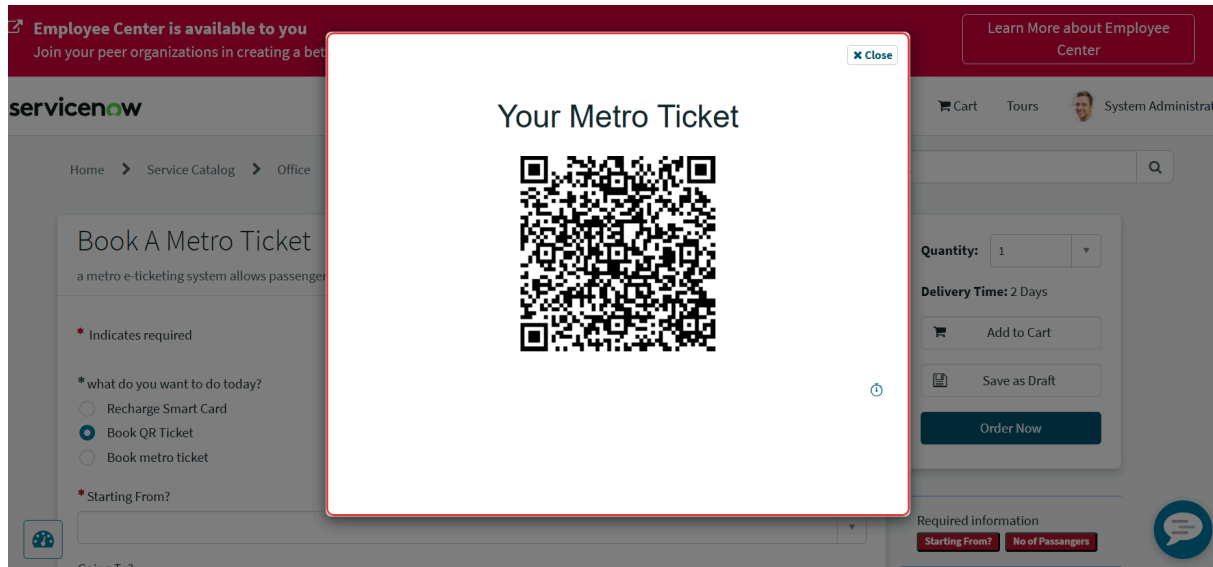
The QR Code Generation feature is used to:

- Generate a unique QR code for every booked metro ticket.
- Provide passengers with a digital ticket for easy access.
- Enable quick and secure ticket verification.
- Eliminate the need for paper tickets.
- Improve passenger convenience and reduce manual verification.

Steps to Generate the QR Code

1. Open the **Service Portal**.
2. Search for **Book a Metro Ticket**.
3. Open the **Book a Metro Ticket** catalog item.
4. Select **Book a QR Ticket** from the available ticket options.
5. Enter all the required journey details:
 - Passenger Name
 - Starting From
 - Going To
 - Journey Type
 - Number of Passengers
 - Travel Date
6. Click **Order Now** to submit the booking request.

7. The system processes the request and generates a unique QR code.
8. The QR code is displayed in a popup (or on a separate page, depending on the implementation).
9. The passenger can scan or save the QR code as the digital metro ticket.
10. Close the QR code window to continue to the request confirmation page.



6.8 Testing the Flow and Verifying Record Creation

After submitting the **Book a Metro Ticket** catalog item, it is important to verify that the **Flow Designer** has executed successfully and that a new booking record has been created in the **Metro Ticket** custom table.

Purpose

The purpose of this testing is to:

- Verify that the Flow Designer is triggered successfully.
- Ensure that the **Get Catalog Variables** action retrieves all passenger and journey details correctly.
- Confirm that the **Create Record** action stores the booking details in the **Metro Ticket** custom table.
- Validate the end-to-end metro ticket booking process.
- Ensure that all booking information is stored accurately without manual intervention.

Steps to Test the Flow

1. Open the **Service Portal**.
2. Search for **Book a Metro Ticket**.
3. Open the catalog item.
4. Enter all the required booking details.
5. Click **Order Now** to submit the booking request.

6. Wait until the request is processed successfully.
7. Open the **Application Navigator**.
8. Navigate to the **Metro Ticket** custom table.
9. Open the **Metro Ticket** records list.
10. Verify that a new record has been created with the submitted booking details.
11. Compare the values in the record with the information entered in the catalog item to ensure data accuracy.

7. Screenshots of Output

1. Ticket generation

2. Record creation

Metro databases Amount Search									
Actions on selected rows... New									
	Amount	Going To	Journey Type	Mode of Payment	no.of passangers	Payment Mode	QR Code	Recharge Amount	Smart Card Name
2	100	(empty)	Single Journey	UPI	3			467	manasa
	60	(empty)	Single Journey	UPI	5			1000	dollar
	100	nagole	Single Journey	UPI	1				
	(empty)	nagole	Single Journey	UPI	1				

8. Conclusion

The Metro Ticket Generation System demonstrates the effectiveness of using the ServiceNow platform to automate and simplify the metro ticket booking process. By utilizing features such as the Service Catalog, Catalog Variables, Catalog Client Scripts, Custom Tables, Flow Designer, and QR Code Generation, the system provides a streamlined workflow that manages the entire ticket booking process efficiently, from ticket request submission to digital ticket generation.

The automation implemented in this project reduces manual intervention, minimizes waiting time at ticket counters, and ensures that every booking follows a standardized workflow. Automatic record creation, fare calculation, QR code generation, and digital ticket management improve booking accuracy, enhance passenger convenience, and provide a secure and paperless ticketing experience.

The system also maintains complete booking records in the custom Metro Ticket table, making it easy to track, manage, and retrieve ticket information whenever required. This centralized approach improves operational efficiency and supports better management of metro ticket bookings.

In conclusion, the Metro Ticket Generation System provides a flexible, scalable, and user-friendly solution for digital metro ticket booking. It enhances the overall passenger experience by offering a fast, reliable, and automated ticketing process. The solution can be further enhanced by integrating features such as online payment gateways, real-time metro schedules, ticket cancellation, refund processing, mobile notifications, and QR code validation at metro entry gates, making it suitable for future smart transportation systems.

experience, and establishes a strong framework that can be extended in the future by integrating additional network services and advanced automation capabilities.